

1. Bias-Variance Decomposition

1. Setup

Regression setting: $y = f(x) + \varepsilon$ where $\mathbb{E}[\varepsilon] = 0$, $\text{Var}(\varepsilon) = \sigma^2$.

Learner: Train on sample S , output $\hat{f}_S(x)$.

Goal: Understand expected prediction error at new point x .

2. Fundamental Decomposition

For squared loss:

$$\mathbb{E}_S \left[\left(y - \hat{f}_S(x) \right)^2 \right] = \text{Bias}^2 + \text{Variance} + \sigma^2$$

where:

- **Bias:** $\mathbb{E}_S [\hat{f}_S(x)] - f(x)$
- **Variance:** $\mathbb{E}_S \left[\left(\hat{f}_S(x) - \mathbb{E}_S [\hat{f}_S(x)] \right)^2 \right]$
- **Noise:** σ^2 (irreducible)

Trade-off: Complex models have low bias but high variance. Simple models have high bias but low variance.

3. Example: Polynomial Regression

Model: $\hat{f}(x) = \sum_{j=0}^k \beta_j x^j$

Degree k small: High bias (underfitting), low variance

Degree k large: Low bias, high variance (overfitting)

Optimal k : Balances bias and variance

2. Ill-Posed Problems

1. Hadamard's Criteria

Problem $Ax = b$ is **well-posed** if:

1. **Existence:** Solution exists for all b
2. **Uniqueness:** Solution is unique
3. **Stability:** Solution depends continuously on data

Ill-posed: Violates one or more criteria.

2. Model Selection as Ill-Posed

Issue: Small changes in training data can lead to drastically different model choices.

Symptom: High variance in selected model complexity across different samples.

Solution: Regularization to stabilize the problem.

3. Tikhonov Regularization

1. General Formulation

Regularized problem: $\min_{h \in \mathcal{H}} [R_{\text{emp}}(h) + \lambda \Omega(h)]$

where:

- $R_{\text{emp}}(h)$: empirical risk (data fit)
- $\Omega(h)$: regularization term (complexity penalty)
- $\lambda > 0$: regularization parameter

Trade-off: λ small $\rightarrow$ low bias, high variance; λ large $\rightarrow$ high bias, low variance

2. Theoretical Guarantees

Regularization as approximation: $R(h_\lambda) \leq \inf_{h \in \mathcal{H}} [R(h) + \lambda \Omega(h)] + O\left(\frac{1}{\sqrt{m}}\right)$

Benefit: Stability (continuous dependence on data) + improved generalization.

3. Choosing λ

Methods:

- **Cross-validation:** Split data, try multiple λ , pick best
- **Hold-out validation:** Reserve validation set
- **Theoretical analysis:** $\lambda = O\left(\frac{1}{\sqrt{m}}\right)$ often optimal
- **Early stopping:** Implicit regularization in gradient descent

4. Ridge Regression

1. Problem Setup

Linear model: $y = X\beta + \varepsilon$

Ordinary Least Squares:

$$\hat{\beta}_{\text{OLS}} = \arg \min_{\beta} \|y - X\beta\|_2^2$$

Issue: High variance when $X^\top X$ is ill-conditioned or singular.

2. Ridge Solution

Ridge regression:

$$\hat{\beta}_{\text{ridge}} = \arg \min_{\beta} [\|y - X\beta\|_2^2 + \lambda \|\beta\|_2^2]$$

Closed form:

$$\hat{\beta}_{\text{ridge}} = (X^\top X + \lambda I)^{-1} X^\top y$$

Effect: Shrinks coefficients toward zero, stabilizes inversion.

3. Properties

Bias: $\mathbb{E}[\hat{\beta}_{\text{ridge}}] \neq \beta$ (biased estimator)

Variance: Lower than OLS, decreases with λ

Mean Squared Error: Often lower than OLS due to bias-variance trade-off

Connection to SVD: $\hat{\beta}_{\text{ridge}} = V D_\lambda U^\top y$ where D_λ shrinks small singular values

4. Bias-Variance Analysis

Let $X = U\Sigma V^\top$ (SVD). Then:

Bias: Increases with λ (shrinkage)

Variance: Decreases with λ :

$$\text{Var}(\hat{\beta}_{\text{ridge}}) = \sigma^2 V (\Sigma^2 + \lambda I)^{-1} \Sigma^2 (\Sigma^2 + \lambda I)^{-1} V^\top$$

Optimal λ : Minimizes total MSE

5. Practical Guidelines

1. Implementation Checklist

1. **Standardize features:** Mean 0, variance 1
2. **Split data:** Train/validation/test
3. **Cross-validation:** Grid search over λ
4. **Evaluate:** Report test error, not training error
5. **Interpretation:** Large coefficients indicate important features

2. When to Use Ridge

Good for:

- High-dimensional data ($p \approx n$ or $p > n$)
- Multicollinearity (correlated features)
- Ill-conditioned design matrix
- Need coefficient shrinkage

Not good for:

- Sparse solutions (use Lasso instead: ℓ_1 penalty)
- Feature selection (Ridge shrinks but doesn't zero out)

6. Related Methods

1. Lasso (L1 Regularization)

Formulation:

$$\min_{\beta} [\|y - X\beta\|_2^2 + \lambda \|\beta\|_1]$$

Effect: Sparse solutions (many $\beta_j = 0$)

Use case: Feature selection, high-dimensional data

2. Elastic Net

Combines Ridge and Lasso:

$$\min_{\beta} [\|y - X\beta\|_2^2 + \lambda_1 \|\beta\|_1 + \lambda_2 \|\beta\|_2^2]$$

Benefits: Sparsity + grouping effect for correlated features

7. Key Pitfalls

Bias-variance for loss, not parameters: Decomposition applies to **prediction error**, not parameter estimates. Low-bias parameter estimate can have high prediction bias.

$\lambda = 0$ is **not always optimal**: Even with infinite data, regularization can help if model is misspecified.

Standardization matters: Ridge is not scale-invariant. Must standardize features first.

Test on held-out data: Never tune λ on test set. Use validation set or cross-validation.

Early stopping $\neq$ explicit regularization: They have similar effects but different theoretical properties.

Ridge doesn't do feature selection: Shrinks toward zero but doesn't set coefficients exactly to zero (unlike Lasso).